Dipendra Sharma

Full Stack AI Engineer Intern | RAG & LLM Systems

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Summary

Full Stack AI Engineer with a strong foundation in backend systems with experience inbuilding RAG pipelines from first principles. Skilled in Python, FastAPI, TypeScript, Next.js, Node.js and PostgreSQL. I've the Experience in document ingestion, chunking strategies, vector search, BM25, and evaluating RAG performance using hit@k and MRR. I've Work Extensive on AI coding assistants like Cursor and Claude Code to improve a coding development while reviewing and improving generated code. Strong focus on shipping production features and integrating LLMs (OpenAI, Ollama) into robust full-stack architectures.

Technical Skills

AI & LLMs: RAG, Local LLMs (Ollama), Qwen, Nomic Embed, LangChain, OpenAI API, Prompt Engineering
Retrieval & Eval: Vector Search, BM25, Hybrid RRF, Chunking Strategies (Fixed, Recursive, Table-Aware), hit@k, MRR
AI Dev Tools: Cursor, Claude Code, Paseo, Git, GitHub Actions
Languages: TypeScript, JavaScript, Python, SQL
Full Stack & Backend: Next.js, React.js, Node.js, Express.js, REST APIs, JWT, RBAC, Jest
Databases & Cloud: PostgreSQL, MongoDB, Supabase, Prisma, Docker, AWS (EC2, S3, IAM), Linux

AI & Full Stack Projects

Chayan | GitHub **Oct 2026 – Present**

- Building a **local-first RAG system** in Python evaluating 12 chunker-retriever combinations. Achieved **hit@5 of 0.89 and MRR of 0.79** using a table-aware chunker with BM25, outperforming hybrid search on complex financial PDFs.
- Deployed a fully local LLM inference pipeline using **Ollama and Qwen 2.5 3B** on a constrained 4GB GPU, eliminating cloud API costs and keeping sensitive document data entirely private.
- Engineered an **LLM relevance gate** (Qwen evaluates top 3 chunks to reject hallucinations) and a **query rewrite step** for temporal normalization (e.g., converting "Q3 2025" into report-specific terminology).
- Developed a custom evaluation harness for multi-page questions and a **fine-tuned adapter matrix** to reshape question embeddings, trained on a strictly non-overlapping dataset to prevent data leakage.
- ***Tech: Python, uv, Ollama, Qwen 2.5, Nomic-Embed, BM25, Hybrid RRF, NumPy, pypdf***

Yeti Jobs | GitHub | Live **Jan 2026 – Apr 2026**

- Built an **AI-powered resume analyzer** using OpenAI and Grok. Users upload a PDF, the system extracts text via `pdf-parse`, sends it to the LLM, and returns a **score (0 to 100)** with actionable ATS feedback.
- Implemented secure file uploads via **Supabase storage** and enforced strict **Zod schema validation** on the LLM's JSON response, preventing frontend crashes from malformed AI output.
- Integrated the AI feature into a **full-stack job portal** (Next.js, React, Node.js, JavaScript) with **50+ REST APIs** and **10 PostgreSQL tables**. Optimized DB queries (**30% faster**) and containerized with Docker (**74% image reduction**).
- Load-tested backend to **67 req/sec** with **0 failures**; maintained **99.5% uptime** over 3 months via cron-based automation.
- ***Tech: OpenAI/Grok, pdf-parse, Next.js, React, Node.js, Express, TypeScript, PostgreSQL, Docker, JWT, Supabase, Zod***

Achievements

- Built an **AI-powered fraud detection system** at a Nepal Police hackathon, using pattern recognition to connect similar cases and identify fraudulent activities (team lead, 24-hour deadline).
- Selected for **GirlScript Summer of Code** 2026.
- Winner of **CodeaThon 7.0** (cash prize, 120+ participants).
- **313 LeetCode** problems solved | **20+ GitHub repos** | **1000+ commits**.
- Shortlisted for **Round 2 – Polaris Fellowship 2026** (selected from thousands across 300+ cities).

Certifications

- **AWS Educate Introduction to Cloud 101** – EC2, S3, IAM fundamentals
- **Claude 101** – Prompt Engineering, Artifacts, Claude Skills

Education

Chitkara University **Aug 2024 – Expected 2027**

Bachelor of Computer Applications (BCA) | CGPA: 9.38/10 *Himachal Pradesh, India*

- Coursework: DSA, Operating Systems, DBMS, Software Engineering, Computer Networking